

Math 67 – HW 8 Solutions

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Chapter 9

C6. We want an orthonormal basis for $\text{null}(T)$, where $T \in \mathcal{L}(\mathbb{C}^4)$ has canonical matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

First we determine a basis for $\text{null}(T)$. For $v = (x, y, z, w)$, we have that $Av = 0$ only if $x + y + z + w = 0$. Then

$$\begin{array}{cccc} x & = & -y & -z & -w \\ y & = & y & & \\ z & = & & z & \\ w & = & & & w \end{array} \Rightarrow \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = y \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + w \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

Then $\{v_1 = (-1, 1, 0, 0), v_2 = (-1, 0, 1, 0), v_3 = (-1, 0, 0, 1)\}$ is a basis for $\text{null}(T)$, but it is not orthonormal. The Gram Schmidt process on $\{v_1, v_2, v_3\}$ yields an orthonormal basis

$$\left\{ \frac{1}{\sqrt{2}}(-1, 1, 0, 0), \quad \frac{1}{\sqrt{6}}(1, 1, -2, 0), \quad \frac{1}{\sqrt{10}}(1, 1, 1, -3) \right\}.$$

P6. Let V be a finite-dimensional inner product space over $\mathbb{F}$ and let U be a subspace of V . Then for the orthogonal complement $U^\perp$ of U ,

$$\dim(U^\perp) = \dim(V) - \dim(U).$$

Proof. Since $V = U \oplus U^\perp$ (Theorem 9.6.2), we have that $V = U + U^\perp$ and $U \cap U^\perp = \{0\}$, and the proof of the problem statement follows from Theorem 5.4.6. $\square$

P8. Let V be a finite-dimensional inner product space over $\mathbb{F}$ and let $P \in \mathcal{L}(V)$ have the properties that (a) $P^2 = P$ and (b) for any $u \in \text{null}(P)$ and $v \in \text{range}(P)$, their inner product $\langle u, v \rangle = 0$. Then P is an orthogonal projection.

Proof. According to the [Wikipedia entry on Projection in linear algebra](#), “An orthogonal projection is a projection for which the range and the null space are orthogonal subspaces.” A linear map P with the property (a) is by definition a projection. We can restate (b) as $\text{null}(P) = (\text{range}(P))^\perp$. Basically, properties (a) and (b) are the definition of an orthogonal projection. $\square$

P9. The claim is **False**. For $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \in \mathbb{C}^{2 \times 2}$, we have $\text{null}(A) = \text{span}\{(-2, 1)\}$ and $\text{range}(A) = \text{span}\{(1, 1)\}$. Then, for $(-2, 1) \in \text{null}(A)$ and $(1, 1) \in \text{range}(A)$, observe that $\langle (-2, 1), (1, 1) \rangle = -1 \neq 0$, and thus $\text{null}(A) \neq (\text{range}(A))^\perp$.

How do we know $(1, 1) \in \text{range}(A)$? Because for any $(x, y) \in \mathbb{C}^2$,

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

Chapter 10

C1. The change of basis transformation S for the canonical basis $e = (e_1, e_2, e_3)$ to $f = (f_1, f_2, f_3)$ of $\mathbb{R}^3$, where

$$f_1 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad f_2 = \frac{1}{\sqrt{6}}(1, -2, 1), \quad f_3 = \frac{1}{\sqrt{2}}(1, 0, -1)$$

$$\text{is } S = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix}.$$

Note that since we are working in $\mathbb{F}^3$, we can write this in matrix notation as

$$S = f^H e = \begin{bmatrix} f_1^H \\ f_2^H \\ f_3^H \end{bmatrix} [e_1 \quad e_2 \quad e_3] = f^H,$$

since

$$e = [e_1 \quad e_2 \quad e_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

The notation $(\cdot)^H$ denotes Hermitian (conjugate) transpose.

C2. For $v = (1, i, -1, -i) \in \mathbb{C}^4$, we want the matrix (with respect to the canonical basis) of the orthogonal projection P such that $\text{null}(P) = \{v\}^\perp$.

First, we need to determine a formulation for P . It is an orthogonal projection, so its range must be the orthogonal complement of its nullspace. Then $\text{range}(P) = (\{v\}^\perp)^\perp = \text{span}\{v\}$, i.e. P maps any vector $x \in \mathbb{C}^4$ to a multiple of v . For $x \in \mathbb{C}^4$, let

$$P(x) = \frac{\langle x, v \rangle}{\|v\|^2} v.$$

This is the component of x in the direction of v , otherwise known as the orthogonal projection of x onto v . Note that

$$P^2 x = P(P(x)) = \frac{\left\langle \frac{\langle x, v \rangle}{\|v\|^2} v, v \right\rangle}{\|v\|^2} v = \frac{\frac{\langle x, v \rangle}{\|v\|^2} \langle v, v \rangle}{\|v\|^2} v = \frac{\langle x, v \rangle}{\|v\|^2} v = Px.$$

The matrix of P with respect to an orthonormal basis (e_1, e_2, e_3, e_4) for $\mathbb{C}^4$ is given as $A = [a_{ij}]$, where $a_{ij} = \langle Pe_j, e_i \rangle$. In the case of the canonical basis $\{e_1 = (1, 0, 0, 0), e_2 = (0, 1, 0, 0), e_3 = (0, 0, 1, 0), e_4 = (0, 0, 0, 1)\}$, it is simply

$$[Pe_1 \ Pe_2 \ Pe_3 \ Pe_4] = \frac{1}{\|v\|^2} \begin{bmatrix} v & -iv & -v & iv \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & -i & -1 & -i \\ i & 1 & -i & -1 \\ -1 & i & 1 & -i \\ -i & -1 & i & 1 \end{bmatrix}.$$

P2. Let V be a finite-dimensional vector space over $\mathbb{F}$ and suppose that $T \in \mathcal{L}(V)$ has the same matrix of transformation with respect to any basis of V . Then T is a multiple of the identity map on V .

Proof. For the zero map $T = 0 \in \mathcal{L}(V)$ this is trivial, since $T = 0 \cdot I_V$. Thus, we prove the case where $T \neq 0$. By Theorem 7.4.1, T has an eigenvalue in $\mathbb{F}$ which we will call α , and an associated eigenvector $v \in V$ such that $T(v) = \alpha v$. Let $n = \dim(V)$, and let $A \in \mathbb{F}^{n \times n}$ be the matrix of transformation for T , which is the same with respect to any basis of V . Then $Ax = \alpha x$, where $x = [v]_b \in \mathbb{F}^n$ for some arbitrary basis $b = (b_1, b_2, \dots, b_n)$. Now let y be any vector in $\mathbb{F}^n$, and let $S \in \mathbb{F}^{n \times n}$ be an invertible matrix such that $x = Sy$ (and $S^{-1}x = y$).¹ Then $y = [v]_c$, for the basis $c = (c_1, c_2, \dots, c_n)$ where $Sc_j = b_j$ and S is the change of basis transformation from c to b . Since A remains the same under any change of basis, we must have $A = SAS^{-1}$. Then

$$SAy = SAS^{-1}x = Ax = \alpha x,$$

and then

$$Ay = \alpha S^{-1}x = \alpha y.$$

Thus, we see that any $y \in \mathbb{F}^n$ is an eigenvector of A , with eigenvalue α . Then it follows (see chapter 7 proof problem 4 from hw#5) that $A = \alpha I \in \mathbb{F}^{n \times n}$, which implies that $T = \alpha I_V \in \mathcal{L}(V)$. $\square$

¹One way to construct such a matrix is as follows: Since the choice of basis b is arbitrary, let $\{b_1, b_2, \dots, b_{n-1}\} \subset \mathbb{F}^n$ be a linearly independent set such that each $b_j \in \mathbb{F}^n$ is not a multiple of x . Then $b = (x, b_1, b_2, \dots, b_{n-1})$ (orthonormalized via Gram-Schmit process) is a basis for $\mathbb{F}^n$. Similarly, let $\{c_1, c_2, \dots, c_{n-1}\} \subset \mathbb{F}^n$ be a linearly independent set that is also linearly independent with y . Then $c = (y, c_1, c_2, \dots, c_{n-1})$ is also a basis for $\mathbb{F}^n$. Then, define S as the change of basis transformation from c to b via the method described in the text. Note that y maps to x under S .